

User Story: US-001 – Student Registration		
Scenario ID	API Test Scenario	Type
ATS-001	Verify student registration with valid Name, Email, Mobile Number, and Password.	Positive
ATS-002	Verify registration fails when mandatory fields are empty.	Validation
ATS-003	Verify registration with invalid email format.	Validation
ATS-004	Verify registration with duplicate email address.	Validation
ATS-005	Verify registration with invalid mobile number.	Validation
ATS-006	Verify password complies with the defined password policy.	Validation
ATS-007	Verify registration with password below minimum or above maximum length.	Boundary
ATS-008	Verify registration with null or invalid input values.	Negative
ATS-009	Verify successful registration creates a student account.	Functional
ATS-010	Verify successful registration returns a Student ID in the response.	Response Validation
ATS-011	Verify the Registration API accepts only the POST method.	API Validation
ATS-012	Verify the Registration API with the required Content-Type: application/json header.	Header Validation
ATS-013	Verify the API returns the expected HTTP status codes for successful and failed registration requests.	Status Code
ATS-014	Verify the response body contains the expected success or error message.	Response Validation
ATS-015	Verify the Registration API responds within the acceptable response time.	Performance
ATS-016	Verify the Registration API handles special characters and SQL injection attempts safely.	Security

User Story: US-002 – Student Login		
Scenario ID	API Test Scenario	Type
ATS-001	Verify student can log in with valid credentials.	Positive
ATS-002	Verify login fails with invalid credentials.	Negative
ATS-003	Verify login fails when email or password is empty.	Validation
ATS-004	Verify email and password validation rules.	Validation
ATS-005	Verify only active users can log in.	Functional
ATS-006	Verify inactive users cannot log in.	Functional
ATS-007	Verify successful login redirects to the dashboard.	Functional
ATS-008	Verify successful login generates an authentication token.	Authentication
ATS-009	Verify failed login does not generate an authentication token.	Authentication
ATS-010	Verify protected APIs require a valid authentication token.	Authorization
ATS-011	Verify direct access to protected resources without a token is denied.	Authorization
ATS-012	Verify invalid login returns the appropriate error message and HTTP status code.	Response Validation
ATS-013	Verify Login API accepts only the POST method.	API Validation
ATS-014	Verify Login API with the required Content-Type: application/json header.	Header Validation
ATS-015	Verify the Login API responds within the acceptable response time.	Performance
ATS-016	Verify the Login API handles SQL injection attempts safely.	Security

User Story: US-003 – Forgot Password		
Scenario ID	API Test Scenario	Type
ATS-001	Verify a registered email can request a password reset.	Positive
ATS-002	Verify an unregistered email cannot request a password reset.	Negative
ATS-003	Verify a password reset OTP/link is sent to the registered email.	Functional
ATS-004	Verify a password reset OTP/link is not sent to an unregistered email.	Negative
ATS-005	Verify an appropriate error message is returned for an unregistered email.	Response Validation
ATS-006	Verify the password reset OTP/link expires after the configured duration.	Security
ATS-007	Verify the password reset OTP/link cannot be reused after a successful password reset.	Security
ATS-008	Verify a student can create a new password using a valid OTP/link.	Functional
ATS-009	Verify the new password complies with the password policy.	Validation
ATS-010	Verify password length validation (minimum and maximum limits).	Boundary
ATS-011	Verify login with the old password fails after a successful password reset.	Functional
ATS-012	Verify login with the new password succeeds after a successful password reset.	Functional
ATS-013	Verify successful password reset returns the appropriate success message and HTTP status code.	Response Validation
ATS-014	Verify invalid or expired OTP/link returns the appropriate error message and HTTP status code.	Response Validation
ATS-015	Verify the Password Reset API accepts only the POST method.	API Validation
ATS-016	Verify the Password Reset API with the required Content-Type: application/json header.	Header Validation
ATS-017	Verify the Password Reset API responds within the acceptable response time.	Performance

ATS-018	Verify the Password Reset API safely handles SQL injection and special character inputs.	Security
---------	--	----------

User Story: US-004– Search Courses		
Scenario ID	API Test Scenario	Type
ATS-001	Verify courses can be searched using a valid keyword.	Positive
ATS-002	Verify no matching courses are returned for an invalid or non-existing keyword.	Negative
ATS-003	Verify matching courses are returned for a valid search keyword.	Functional
ATS-004	Verify leading and trailing spaces in the search keyword are ignored.	Validation
ATS-005	Verify partial keyword search returns all relevant matching courses.	Functional
ATS-006	Verify the response contains Course ID, Title, Category, and Instructor.	Response Validation
ATS-007	Verify search with a complete keyword followed by unrelated text returns appropriate results.	Validation
ATS-008	Verify "No Courses Found" message is returned when no matches exist.	Response Validation
ATS-009	Verify the Search API accepts only the GET method.	API Validation
ATS-010	Verify the Search API with the required authentication header/token (if applicable).	Header Validation
ATS-011	Verify the Search API returns the appropriate HTTP status codes for successful and unsuccessful searches.	Status Code
ATS-012	Verify the Search API responds within the acceptable response time.	Performance
ATS-013	Verify the Search API safely handles SQL injection attempts in the search keyword.	Security
ATS-014	Verify search with an empty keyword returns the expected response (all courses or validation message, as per API specification).	Boundary
ATS-015	Verify search with a single-character keyword (minimum valid keyword) returns the expected response, if supported by the API.	Boundary

User Story: US-005– Course Enrollment		
Scenario ID	API Test Scenario	Type
ATS-001	Verify student can enroll in an available course.	Positive
ATS-002	Verify enrollment fails for an invalid or unavailable Course ID.	Negative
ATS-003	Verify student cannot enroll in the same course twice.	Validation
ATS-004	Verify only authenticated users can enroll in a course.	Authentication
ATS-005	Verify unauthenticated users cannot enroll in a course.	Authentication
ATS-006	Verify successful enrollment creates an enrollment record.	Functional
ATS-007	Verify the enrolled course appears in the student's enrolled course list.	Functional
ATS-008	Verify unenrolled courses are not displayed in the student's enrolled course list.	Functional
ATS-009	Verify the enrollment response contains the expected success message and response body (e.g., Enrollment ID, Course ID, Student ID).	Response Validation
ATS-010	Verify the Enrollment API returns the appropriate HTTP status codes.	Status Code
ATS-011	Verify the Enrollment API accepts only the POST method.	API Validation
ATS-012	Verify the Enrollment API with the required Authorization and Content-Type headers.	Header Validation
ATS-013	Verify the Enrollment API responds within the acceptable response time.	Performance
ATS-014	Verify the Enrollment API safely handles SQL injection or invalid Course ID input.	Security

User Story: US-006– Student Dashboard		
Scenario ID	API Test Scenario	Type
ATS-001	Verify the Dashboard API accepts only the GET method.	API Validation
ATS-002	Verify the dashboard displays all enrolled courses.	Functional
ATS-003	Verify the dashboard does not display unenrolled courses.	Functional
ATS-004	Verify the dashboard displays course progress.	Functional
ATS-005	Verify the dashboard displays instructor details.	Functional
ATS-006	Verify the dashboard displays the correct completion percentage.	Functional
ATS-007	Verify the completion percentage is calculated correctly.	Validation
ATS-008	Verify the response body contains the expected dashboard details (Course ID, Course Name, Progress, Instructor, Completion Percentage).	Response Validation
ATS-009	Verify the Dashboard API returns the appropriate HTTP status codes.	Status Code
ATS-010	Verify only enrolled students can access the dashboard using a valid authentication token.	Authentication
ATS-011	Verify dashboard access is denied for requests with an invalid or missing authentication token.	Authentication
ATS-012	Verify the Dashboard API responds within the acceptable response time.	Performance
ATS-013	Verify the Dashboard API includes the required request headers (for example, Authorization: Bearer <token>).	Header Validation

User Story: US-007– Track Course Progress		
Scenario ID	API Test Scenario	Type
ATS-001	Verify course progress updates automatically after lesson completion.	Functional
ATS-002	Verify completed lessons are marked as completed.	Functional
ATS-003	Verify incomplete lessons are not marked as completed.	Negative
ATS-004	Verify progress is retained after logout and login.	Functional
ATS-005	Verify course completion reaches 100% after all lessons are completed.	Functional
ATS-006	Verify course completion status is updated correctly.	Functional
ATS-007	Verify the response body contains Course ID, Lesson ID, Progress Percentage, and Completion Status.	Response Validation
ATS-008	Verify the Progress API returns the appropriate HTTP status codes.	Status Code
ATS-009	Verify the Progress API supports the GET method for viewing progress.	API Validation
ATS-010	Verify the Progress Update API supports the PUT/PATCH method for updating progress (as per API specification).	API Validation
ATS-011	Verify the Progress API includes the required Authorization header/token.	Header Validation
ATS-012	Verify unauthenticated users cannot view or update course progress.	Authentication
ATS-013	Verify an invalid Course ID or Lesson ID returns the appropriate error message and HTTP status code.	Negative
ATS-014	Verify the Progress API responds within the acceptable response time.	Performance

User Story: US-008– Download Course Certificate		
Scenario ID	API Test Scenario	Type
ATS-001	Verify the student can successfully download the course completion certificate.	Positive
ATS-002	Verify the certificate cannot be downloaded before course completion.	Negative
ATS-003	Verify the downloaded certificate contains the correct Student Name.	Response Validation
ATS-004	Verify the downloaded certificate contains the correct Course Name.	Response Validation
ATS-005	Verify the downloaded certificate contains the correct Completion Date.	Response Validation
ATS-006	Verify the certificate is generated in PDF format.	Functional
ATS-007	Verify only authenticated users can download their certificate.	Authentication
ATS-008	Verify unauthenticated users cannot download the certificate.	Authentication
ATS-009	Verify a student cannot download another student's certificate.	Authorization
ATS-010	Verify the Certificate Download API returns the appropriate HTTP status codes.	Status Code
ATS-011	Verify the response headers indicate a PDF file (e.g., Content-Type: application/pdf).	Header Validation
ATS-012	Verify the Certificate Download API responds within the acceptable response time.	Performance
ATS-013	Verify an invalid or non-existent certificate request returns an appropriate error message and HTTP status code.	Negative
ATS-014	Verify the Certificate Download API supports the HTTP method specified in the API documentation.	API Validation

User Story: US-009– Create Course (Instructor)		
Scenario ID	API Test Scenario	Type
ATS-001	Verify an instructor can successfully create a new course.	Positive
ATS-002	Verify a student cannot create a course.	Authorization
ATS-003	Verify an unauthenticated user cannot create a course.	Authentication
ATS-004	Verify course creation fails when the Course Title is empty.	Validation
ATS-005	Verify course creation succeeds with an optional Description.	Functional
ATS-006	Verify learning content can be uploaded successfully while creating the course.	Functional
ATS-007	Verify the created course is visible to students after it is published.	Functional
ATS-008	Verify a unique Course ID is generated after successful course creation.	Response Validation
ATS-009	Verify the response body contains the Course ID, Title, Description, and publication status.	Response Validation
ATS-010	Verify the Course Creation API returns the appropriate HTTP status codes.	Status Code
ATS-011	Verify the Course Creation API includes the required request headers (Authorization, Content-Type).	Header Validation
ATS-012	Verify the Course Creation API responds within the acceptable response time.	Performance
ATS-013	Verify course creation with invalid or unsupported learning content is rejected with an appropriate error message.	Negative
ATS-014	Verify the Course Creation API supports the HTTP method specified in the API documentation.	API Validation
ATS-015	Verify the API safely handles malicious input (e. g., SQL injection or script injection) in the Course Title and Description fields.	Security

User Story: US-0010– User Management		
Scenario ID	API Test Scenario	Type
ATS-001	Verify only an Admin can access the User Management API.	Authorization
ATS-002	Verify an Admin can view the list of all users.	Functional
ATS-003	Verify an Admin can successfully activate a deactivated user.	Functional
ATS-004	Verify an Admin can successfully deactivate an active user.	Functional
ATS-005	Verify the user's status is updated immediately after activation.	Functional
ATS-006	Verify the user's status is updated immediately after deactivation.	Functional
ATS-007	Verify a non-admin user cannot access the User Management API.	Authorization
ATS-008	Verify an unauthenticated user cannot access the User Management API.	Authentication
ATS-009	Verify activation of a non-existent User ID returns an appropriate error message.	Negative
ATS-010	Verify deactivation of a non-existent User ID returns an appropriate error message.	Negative
ATS-011	Verify the response body contains the correct user details and updated status after activation/deactivation.	Response Validation
ATS-012	Verify the User Management API returns the appropriate HTTP status codes for all operations.	Status Code
ATS-013	Verify the User Management API includes the required request headers (Authorization, Content-Type).	Header Validation
ATS-014	Verify the User Management API responds within the acceptable response time.	Performance
ATS-015	Verify the User Management API supports the HTTP methods specified in the API documentation for viewing and updating users.	API Validation
ATS-016	Verify the API safely handles invalid or malicious input (e.g., SQL injection) in User ID or request parameters.	Security

User Story: US-0011– Update Profile		
Scenario ID	API Test Scenario	Type
ATS-001	Verify an authenticated student can update their profile successfully.	Positive
ATS-002	Verify Name and Mobile Number can be updated successfully.	Functional
ATS-003	Verify Email update follows the system's verification rules.	Validation
ATS-004	Verify mandatory fields cannot be left blank.	Validation
ATS-005	Verify updated profile information is reflected immediately.	Functional
ATS-006	Verify the updated profile details are returned in the API response.	Response Validation
ATS-007	Verify an unauthenticated user cannot update the profile.	Authentication
ATS-008	Verify invalid profile data (e.g., invalid email or mobile number) is rejected.	Negative
ATS-009	Verify the Profile Update API returns the appropriate HTTP status codes.	Status Code
ATS-010	Verify the Profile Update API includes the required Authorization and request headers.	Header Validation
ATS-011	Verify the Profile Update API responds within the acceptable response time.	Performance
ATS-012	Verify the Profile Update API supports the HTTP method specified in the API documentation.	API Validation
ATS-013	Verify the API safely handles invalid or malicious input in profile fields.	Security

User Story: US-0012– Logout		
Scenario ID	API Test Scenario	Type
ATS-001	Verify an authenticated student can log out successfully.	Positive
ATS-002	Verify the authentication token/session is invalidated after logout.	Authentication
ATS-003	Verify protected APIs cannot be accessed after logout using the same token/session.	Security
ATS-004	Verify the Logout API returns a successful response after logout.	Response Validation
ATS-005	Verify the student can log in again after successfully logging out.	Functional
ATS-006	Verify an unauthenticated or invalid token request cannot log out.	Negative
ATS-007	Verify the Logout API returns the appropriate HTTP status codes.	Status Code
ATS-008	Verify the Logout API includes the required Authorization/request headers.	Header Validation
ATS-009	Verify the Logout API responds within the acceptable response time.	Performance
ATS-010	Verify the Logout API supports the HTTP method specified in the API documentation.	API Validation

User Story: US-0013– Submit Assignment		
Scenario ID	API Test Scenario	Type
ATS-001	Verify an enrolled student can successfully submit an assignment.	Positive
ATS-002	Verify a non-enrolled student cannot submit an assignment.	Authorization
ATS-003	Verify supported file types are accepted.	Validation
ATS-004	Verify unsupported file types are rejected.	Negative
ATS-005	Verify assignment upload fails when the file size exceeds the maximum limit.	Boundary/Validation
ATS-006	Verify a unique Assignment ID is generated after successful submission.	Response Validation
ATS-007	Verify a successful upload confirmation is returned after submission.	Response Validation
ATS-008	Verify the submitted assignment appears in the student's submissions list.	Functional
ATS-009	Verify an unauthenticated user cannot submit an assignment.	Authentication
ATS-010	Verify the Assignment API returns the appropriate HTTP status codes.	Status Code
ATS-011	Verify the Assignment API includes the required Authorization and request headers.	Header Validation
ATS-012	Verify the Assignment API responds within the acceptable response time.	Performance
ATS-013	Verify the Assignment API supports the HTTP method specified in the API documentation.	API Validation

User Story: US-0014– Attempt Quiz		
Scenario ID	API Test Scenario	Type
ATS-001	Verify an enrolled student can attempt a quiz.	Positive
ATS-002	Verify a non-enrolled student cannot attempt a quiz.	Authorization
ATS-003	Verify quiz answers are submitted successfully.	Functional
ATS-004	Verify the quiz score is calculated correctly after submission.	Functional
ATS-005	Verify the quiz result is displayed after successful submission.	Functional
ATS-006	Verify the maximum quiz attempt limit is enforced.	Validation
ATS-007	Verify the quiz submission time is recorded correctly.	Functional
ATS-008	Verify the quiz result is stored successfully.	Response Validation
ATS-009	Verify an unauthenticated user cannot attempt or submit a quiz.	Authentication
ATS-010	Verify the Quiz API returns the appropriate HTTP status codes.	Status Code
ATS-011	Verify the Quiz API includes the required Authorization and request headers.	Header Validation
ATS-012	Verify the Quiz API responds within the acceptable response time.	Performance
ATS-013	Verify the Quiz API supports the HTTP method specified in the API documentation.	API Validation